PROJECT-BASED LEARNING

PHASE-I EVALUATION

Disaster Management System

Team ID: DSCPP-III-2026-T058

Department of Computer Science & Engineering
Graphic Era (Deemed to be University), Dehradun
Academic Session 2026–27

Team & Mentor Details

Team Information

Team ID: DSCPP-III-2026-T058
Semester: 3rd
Section: B, E, F, G
Domain: *Data Structures and C++*

Team Members

1. *Avika Mittal – 2510012768*
2. *Gaurav Singh Negi – 2510012516*
3. *Avni Khandelwal – 25103140050*
4. *Aryan Gautam – 2510012654*

Mentor

Mr. Navneet Rajpoot

Interactions completed:2

Problem Statement & Motivation

Problem Statement

Managing information about affected people,rescue teams,resources,emergency requests,and locations becomes difficult and unorganized.Our project solves this problem by providing a system to store,update,search,and manage disaster-related information efficiently.

Motivation

- *Disaster data is scattered and unorganized.*
- *Delays effective disaster response.*
- *Existing systems are complex.*

Target Users / Stakeholders

- *Administrators management staff.*
- *Rescue teams and government agencies.*
- *Organized information and efficient response.*

Objectives

Project Objectives

1. *Develop a solution to efficiently manage disaster-related information.*
2. *Implement core functions for managing disaster data.*
3. *Integrate DSA and OOP concepts using C++.*
4. *Evaluate the system using test cases and different disaster scenarios.*
5. *Demonstrate practical applicability and identify future enhancements.*

Key Objective: *Develop a functional C++ system that efficiently manages and processes disaster-related information.*

Proposed Solution

Proposed Approach

1. *Collect disaster-related data.*
2. *Apply DSA concepts.*
3. *Manage, search and update data.*
4. *Use file handling.*
5. *Display results in console.*

Key Features

- *Disaster Records*
- *People Management*
- *Resource Management*
- *Rescue Teams*
- *Location Mapping*

The proposed solution organizes disaster data in one system and uses DSA for efficient management, searching, and processing. It helps improve coordination of people, resources, rescue teams, and locations.

Integration of PBL Course Concepts

Course	Concepts Applied	Project Module
Data Structures in C	*Arrays, LinkedLists, Stack, Queue, Graph, Searching, File Handling*	*Modules: Rescue Team, Request, Route and Location*
OOPs with C++	*Classes, Objects, Functions, Inheritance, Encapsulation*	*Modules: Disaster, Resource*

Requirement: "DSA manages and processes disaster data, while C++ OOP organizes these functions into classes and objects."

System Architecture / Workflow



Major Components

- *Disaster Management Module*
- *Resource and Rescue Module*
- *Location Management Module*

Data / Control Flow

- *User Input→Main Controller→Modules→Data Structures→ Output*
- *Disaster/People/Resources/Requests /Routes Modules*
- *Organize records and display results.*

Technology Stack & Methodology

Technology Stack

- Programming: *C++/ C*
- Database: *File-based Storage (.txt)*
- Tools / Framework: *C++ STL, fstream*
- IDE: *VS Code/ Code::Blocks*
- Version Control: *Git / GitHub*

Development Methodology

1. Identify requirements.
2. Design system structure.
3. Develop using C++ and DSA.
4. Test system functions.
5. Evaluate results.

C++ is selected for easy implementation of DSA and OOP concepts, while the step-by-step methodology ensures systematic development, testing, and evaluation of the system.

Initial Progress & Team Contribution

Progress Achieved

- *Project idea and requirements identified.*
- *DSA and C++ concepts identified.*
- *system architecture and workflow designed.*
- *Basic disaster module implemented.*
- *Initial disaster data prepared.*

Individual Contribution

Member	Role	Contribution
Avika	*Lead*	*Project and design planning*
Avni	*Developer*	*Implementing array,linked list,stack,queue*
Aryan	*DB/Testing*	*Filehandling,testing and debugging*
Gaurav	*Developer/Docs*	*Classes Development and modules*

Roadmap, Expected Outcomes & References

Roadmap

1. Phase-I: Proposal & Design
2. Phase-II: Development
3. Phase-III: Final Implementation

Expected Outcomes

- *Functional C++ system.*
- *Efficient data management.*
- *Supports disaster management.*
- *Easy to expand with new features.*

References

1. *NDMA Disaster Management Resources*
2. *Introduction to Algorithms (CLRS)*
3. *C++ Reference (cppreference)*
4. *GeeksforGeeks Data Structures Examples*

Thank You
Questions & Suggestions